

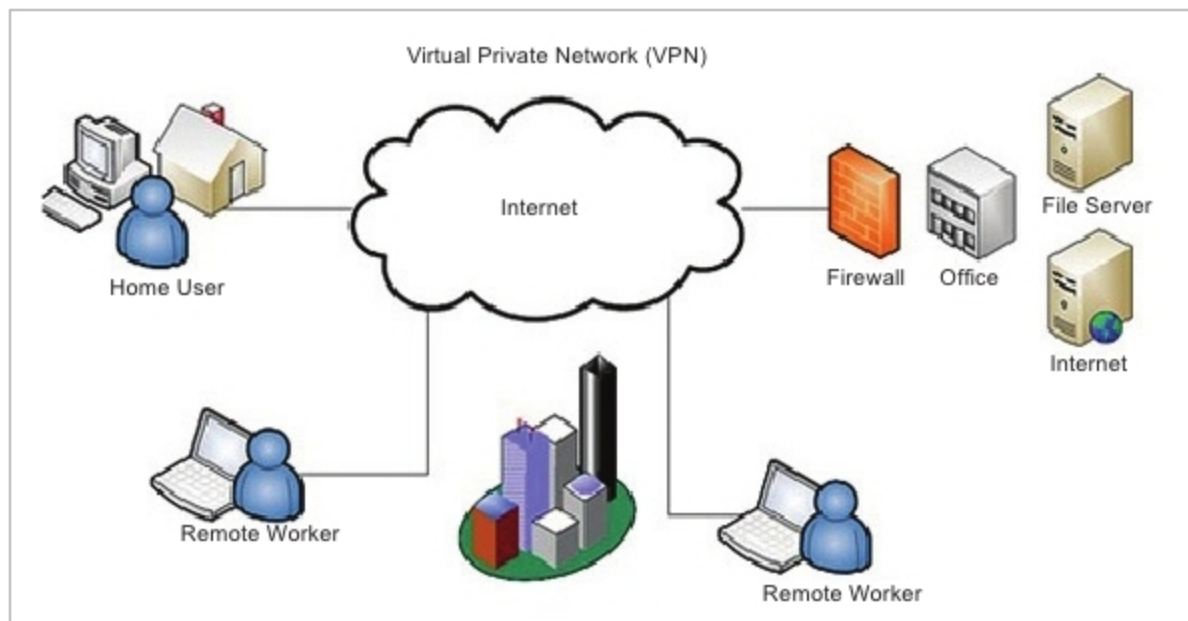


Figure 1: Basics of a VPN

offices in multiple fixed locations to establish secure connections with one another over public networks such as the Internet. A site-to-site VPN extends the company's network, making computer resources from one location available to employees at another. It is also called router-to-router VPN.

Site-to-site VPNs are categorised into the following two types.

- **Intranet based:** If a company has one or more remote locations and wishes to connect them under a single private network, an intranet VPN can be created to connect each separate LAN to a single WAN.
- **Extranet based:** IT organisations working in close collaboration with other organisations can build an extranet VPN that connects to those other organisations' LANs. An extranet VPN allows the companies to work together in a secure, shared network environment while preventing access to separate intranets.

Remote access VPN: A remote access VPN allows a user to connect to a private network and access its services and resources remotely. The connection between the user and the private network happens via the Internet and the connection is secure. There are two main components in remote access VPNs. The first is the network access server (NAS), also termed the media gateway or remote access service (RAS). It is regarded

as a dedicated server in which the user connects to the Internet to use the VPN. The NAS connectivity requires a valid user name and password, and to authenticate the user every NAS has its own authentication procedure. Another requirement is remote access VPN-client software. All the users who want to use VPN services have to install software in the users' own computers to establish and maintain the VPN connection. A remote access VPN is useful both for business as well as home users.

VPN protocols

A VPN primarily depends on how it is used and how much security, performance and uptime is required by specific users. A VPN has many protocols, which enable users to communicate between sites, cloud based vendors, streaming services and even gaming servers.

The following are the basic protocols for VPNs.

- **Point to Point Tunnelling Protocol (PPTP):** This is one of the most common, easy to configure and fast VPN protocols. It is very useful for audio and video streaming, and works without any hiccups on even old devices with limited processing capabilities. PPTP has a lot of security vulnerabilities — its underlying protocols like MS-CHAP-v1/v2 are insecure and can easily be cracked by hackers. It uses 128-bit encryption keys for sensitive data transfers.
- **Layer 2 Tunnelling Protocol (L2TP):** This is a standard protocol for tunnelling L2 traffic over the IP network. Its ability to carry almost any L2 data format over IP or other L3 networks makes it highly useful. The UDP port is used for L2TP communications, but it doesn't provide any security for data like encryption and confidentiality like the IPsec protocol. It consists of two components — the tunnel and the session. The tunnel provides reliable transport between two L2TP Control Connection Endpoints (LCCs) and carries only control packets. The session is logically contained within the tunnel and carries user data. It is not secure by itself and uses IPsec and 3DES encryption methods.
- **IPsec (Internet Protocol Security):** IPsec is a trusted encryption and tunnelling protocol that uses encryption on the IP traffic over a given tunnel. It defines cryptographic algorithms used to encrypt, decrypt and authenticate packets as well as the protocols required for secure key exchange and key management. It contains two mechanisms for securing IP packets. The first is ESP (Encapsulating Security Payload) Protocol, which defines a method for encrypting data in IP packets and the authentication header (AH) for the digital signing of IP packets. The second mechanism to secure IP packets is IKE (Internet Key Exchange), which is used to manage cryptographic keys used by hosts for IPsec.
- **Secure Socket Tunnelling Protocol (SSTP):** This is considered as a top-grade VPN secure protocol with 2048-bit encryption providing a mechanism to transport PPP traffic via the SSL/TLS channel. SSL/TLS provides transport-level security with key negotiation, encryption and traffic integrity checking. SSTP can be used to pass through proxy servers and firewalls. During the session, clients and servers using SSTP must be authenticated during the SSL phase of transmission.